

Chapter 107

LUT Nodes

This chapter details the LUT nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn/MediaOut nodes in DaVinci Resolve are interchangeable with Loader/Saver nodes in Fusion Studio, unless otherwise noted.

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File LUT [FLU]



The File LUT node

File LUT Node Introduction

The File LUT node applies a Lookup table (LUT) to the image: either a simple 1D LUT or a supported 3D LUT. Unlike the Color Curves node, it does not use a spline-based LUT. Instead, it loads the LUT from a file stored on your computer or server.

This approach has two advantages. The first is that the only part of the LUT stored in the composition is the path to the file. Since LUT files can be large, this can dramatically reduce the file size of a composition when several LUTs are present. The second advantage is that it becomes possible to adjust all File LUT nodes using the same file at the same time, just by changing the contents of the LUT. This can be useful when the same LUT-based color correction is applied in many different compositions.

Inputs

The File LUT node includes two inputs: one for the main image and the other for an effect mask to limit the area where the LUT is applied.

Input: This orange input is the only required connection. It accepts a 2D image output that gets the LUT applied.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the applied LUT to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

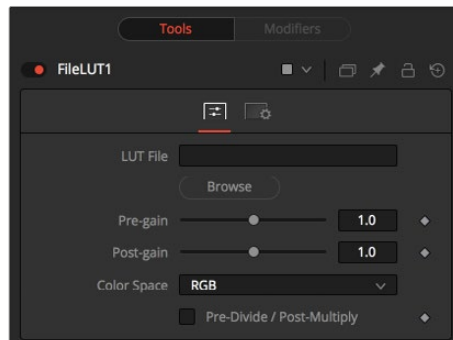
Basic Node Setup

The File LUT node can be placed after a MediaIn node in DaVinci Resolve or a Loader node in Fusion Studio. Sometimes this setup is used to convert the camera-original image to linear color space for compositing. Other times, as in the example below, the File LUT is applied at the end as a grading LUT to apply a look from the colorist.



A File LUT node applied at the end of a node tree as a colorist's look

Inspector



File LUT controls

Controls Tab

The Controls tab includes options for loading a LUT and making adjustments to the gain, color space, and Alpha channel, if one exists.

LUT File

This field is used to enter the path to the LUT file. Clicking the Browse button opens a file browser window to locate the LUT file instead of entering it manually into the LUT File field. Currently, this node supports LUTs exported from Fusion in .LUT and .ALUT formats, DaVinci Resolve's .CUBE format, and several 3D LUT formats. The node fails with an error message on the Console if it is unable to find or load the specified file.

Pre-Gain:

This slider is a gain adjustment before the LUT being applied. This can be useful for pulling in highlights before the LUT clips them.

Post-Gain

This slider is a gain adjustment after the LUT is applied.

Color Space

This menu is used to change the color space the LUT is applied in. The default is to apply the curves described in the LUT to the RGB color space, but options for YUV, HLS, HSV, and others are also available.

Pre-Divide/Post-Multiply

Selecting the Pre-Divide/Post-Multiply checkbox causes the image pixel values to be divided by the Alpha values before applying the LUT, and then re-multiplied by the Alpha value after the correction.

This helps to prevent the creation of illegally additive images, particularly around the edges of a blue/green key or when working with 3D-rendered objects.

Settings Tab

The Settings tab in the Inspector is also duplicated in other LUT nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

LUT Cube Analyzer [LCA]



The LUT Cube Analyzer node

LUT Cube Analyzer Node Introduction

The LUT Cube Analyzer takes an image originated by the LUT Cube Creator as an input and allows the user to create a 3D LUT file in ALUT3, ITX, or 3DL format.

Feeding the original LUT Cube Creator image into the node results in an unaltered, or 1:1, LUT file, and nothing is displayed in the viewer.

You can, however, modify, grade, and color correct the original cube image with as many nodes as you like and feed the result into the LUT Cube Analyzer. This creates a LUT that exactly resembles your color pipeline.

Inputs

The LUT Cube Analyzer includes a single orange input.

Input: The orange input is used to take the output of any node modifying an image that originated with the LUT Cube Creator.

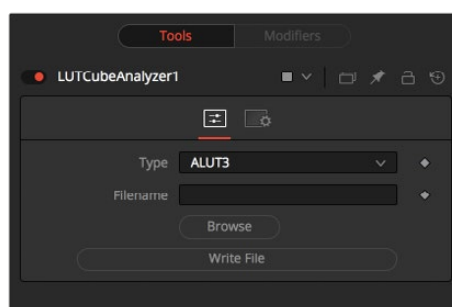
Basic Node Setup

The example below shows a node tree starting with a LUT Cube Creator node and going through two color adjustments. The adjusted image is then connected to a LUT Cube Analyzer, which generates the LUT file.



Generating a LUT starts with the LUT Cube Creator and ends with a LUT Cube Analyzer.

Inspector



LUT Cube Analyzer controls

Controls Tab

The Controls tab for the LUT Cube Analyzer node is used to select the desired LUT output format, specify a filename, and write the 3D LUT to disk.

Type

Select the desired output format of the 3D LUT.

Filename

Enter the path where you want the file saved and enter the name of the LUT file. Alternatively, you can click the Browse button to open a file browser to select the location and filename.

Write File

Press this button to generate the 3D LUT file based on the settings above.

Settings Tab

The Settings tab in the Inspector is also duplicated in other LUT nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

LUT Cube Apply [LCP]



The LUT Cube Apply node

LUT Cube Apply Node Introduction

The LUT Cube Apply node takes an image created by the LUT Cube Creator as the foreground input and applies that LUT to the image connected to the background input.

Feeding the original image into the node would result in an unaltered, or 1:1, output.

You can, however, modify, grade, and color correct the original cube image with as many nodes as you like and feed the result into the LUT Cube Apply. Or, take a LUT image that has been graded beforehand to apply the LUT without having to write an actual 3D LUT using the LUT Cube Analyzer.

Inputs

The LUT Cube Apply has three inputs: a green input where the output of the LUT Cube Creator is connected, an orange input for the image to have the LUT applied, and a blue effect mask input

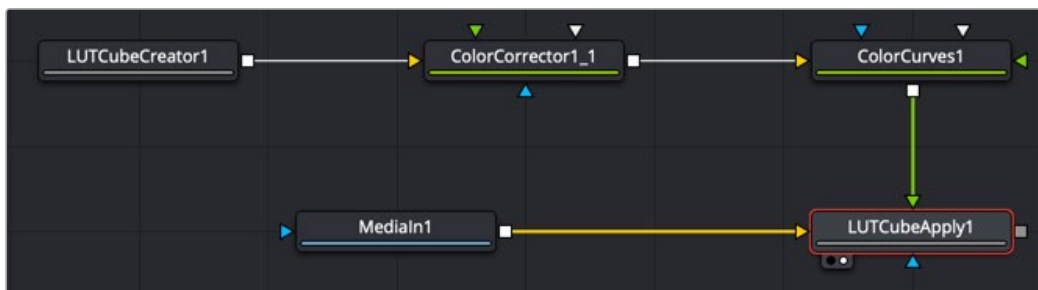
Input: This orange input accepts a 2D image that gets the LUT applied.

Reference Image: The green input is used to connect the output of the LUT Cube Creator or a node that is modifying the image originating in the LUT Cube Creator.

Effect Mask: The optional effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the LUT Cube Apply to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The example below shows a node tree starting with a LUT Cube Creator node and going through two color adjustments. The adjusted image is then connected to the green Reference input of the LUT Cube Apply. The image you want to apply the LUT to is connected to the orange Input.



The LUT generated by the LUT Cube Creator is applied to an image using the LUT Cube Apply node.

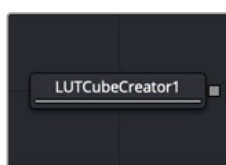
Inspector

There are no controls for the LUT Cube Apply node. The LUT connected to the green foreground input is applied to the image connected to the orange background input without having to write an actual 3D LUT using the LUT Cube Analyzer.

Settings Tab

The Settings tab in the Inspector is also duplicated in other LUT nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

LUT Cube Creator [LCC]



The LUT Cube Creator node

LUT Cube Creator Node Introduction

The LUT Cube Creator generates an image for further use with the LUT Cube Analyzer or LUT Cube Apply nodes. The output can be graded, color corrected, or modified with any tool inside and outside of Fusion. If working outside Fusion, make sure to keep the image in 32-bit floating point to preserve color accuracy.

Feeding the original LUT Cube Creator image into the LUT Cube Analyzer node results in an unaltered, or 1:1, LUT file, and nothing is displayed in the viewer.

Inputs

There are no inputs on the LUT Cube Creator. The purpose of the node is to generate an image that can be used to create a LUT.

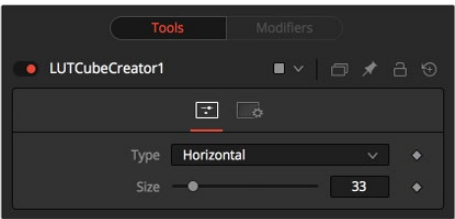
Basic Node Setup

The example below shows a node tree starting with a LUT Cube Creator node and going through two color adjustments. The adjusted image is then connected into a LUT Cube Analyzer that generates the LUT file.



Generating a LUT starts with the LUT Cube Creator and ends with a LUT Cube Analyzer.

Inspector



LUT Cube Creator controls

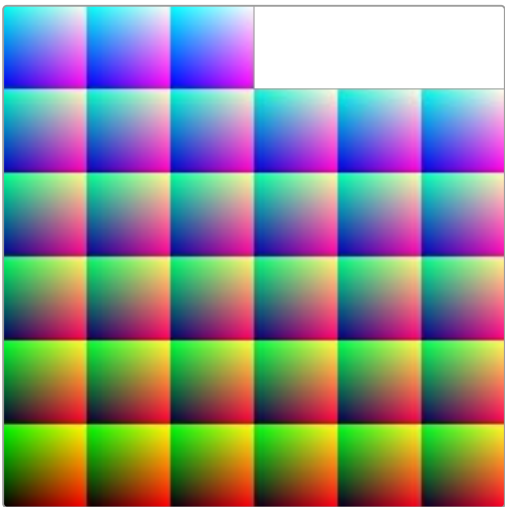
Controls Tab

The Controls tab creates a test pattern of sorts used to create a 3D LUT. The controls here determine the complexity of the pattern used to create a LUT using the LUT Cube Analyzer.

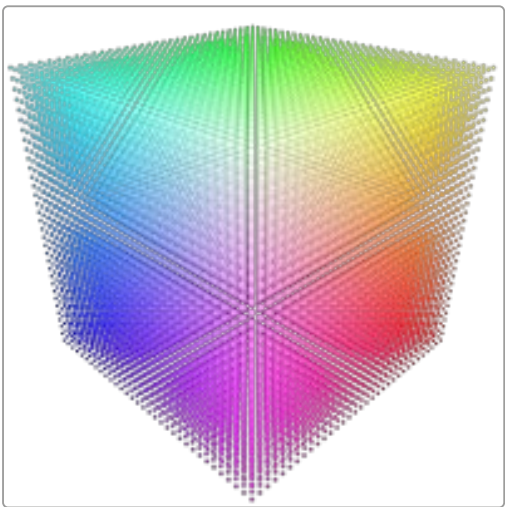
Type:

The Type menu is used to create a pattern of color cubes.

- **Horizontal:** Creates a long, horizontal strip representing a color cube.
- **Vertical:** Creates a long, vertical strip representing a color cube.
- **Rect:** Creates a rectangular image, as depicted below, representing a color cube.



A Cube image created with the Rect type



The resulting color cube

Size

Determines the resolution of the color cube.

Typical Size settings for color cubes are 33 (33 x 33 x 33) or 65 (65 x 65 x 65). These numbers are the samples on each side of the cube. A 33 x 33 x 33 cube has around 35,937 color samples.

NOTE: Higher resolutions yield more accurate results but are also more memory and computationally expensive.

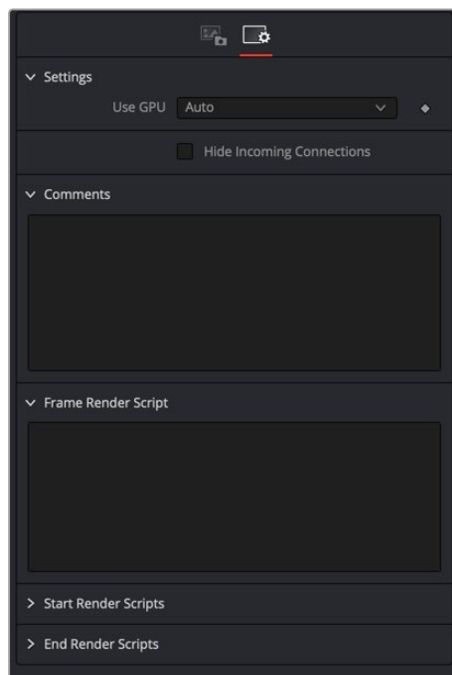
Settings Tab

The Settings tab in the Inspector is also duplicated in other LUT nodes. These common controls are described in the following “The Common Controls” section.

The Common Controls

LUT nodes share a number of identical controls in the Inspector. This section describes controls that are common among LUT nodes.

Inspector



Common LUT Settings Inspector

Settings Tab

The Settings tab in the Inspector can be found on every tool in the LUT category. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options, which are also covered here.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node are displayed in the Inspector. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.